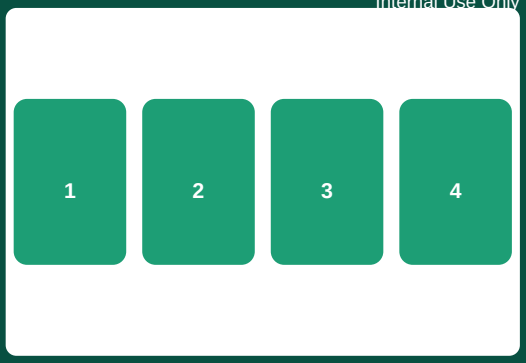
QUALITY & PROCESS

12 Step Kaizen

Run a focused 3–5 day improvement event using a structured 12-step methodology to achieve breakthrough results quickly



WHEN TO USE

When a targeted area needs rapid, measurable improvement with cross-functional team involvement and visible results.

HOW TO APPLY — STEPS

1. Select theme and set specific targets
2. Form and train the team
3. Observe the current state at gemba
4. Collect and analyse process data
5. Generate improvement ideas as a team
6. Select and plan improvements
7. Implement changes during the event
8. Measure results against targets
9. Standardise the new method
10. Train all affected personnel
11. Report to management and stakeholders
12. Celebrate and plan the next improvement event

PRACTITIONER TIP

Implement on Day 3 — spend Day 4 measuring results. Never leave implementation as a post-event action item.

KEY OUTPUTS

- Kaizen event report with before/after metrics
- Updated work instructions
- 30/60/90-day sustain audit plan

HOW TO MEASURE SUCCESS

% of target achieved during event; improvement sustained at 30-day audit

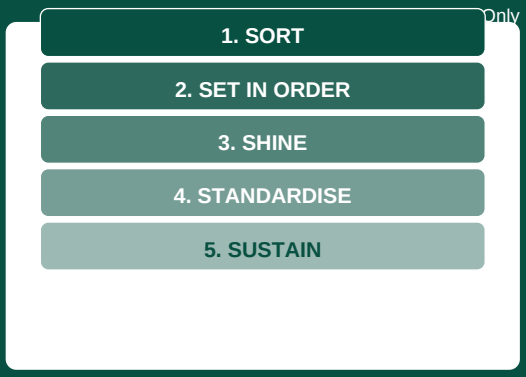
TOOL CATEGORY

Quality & Process

QUALITY & PROCESS

5S

Create and sustain a clean, organised, standardised workplace through Sort, Set in Order, Shine, Standardise, and



WHEN TO USE

When workplaces are cluttered, items cannot be found quickly, or quality and safety issues trace back to the environment.

HOW TO APPLY — STEPS

1. Sort — remove everything not needed for current work
2. Set in order — a defined place for every item
3. Shine — clean and inspect simultaneously every day
4. Standardise — create and display visual standards
5. Sustain — audit regularly and make discipline a habit

PRACTITIONER TIP

! 5S is not a one-off clean-up event. Without Sustain, the workplace reverts within three months.

KEY OUTPUTS

- Before/after photo evidence
- 5S audit checklist and score
- Visual workplace standards

HOW TO MEASURE SUCCESS

5S audit score (0–100); average time to find any item (target under 30 seconds)

TOOL CATEGORY

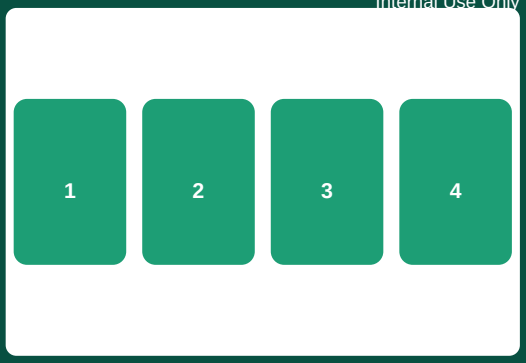
Quality & Process

QUALITY & PROCESS

Internal Use Only

5 Why's

Drill down to root cause by repeatedly asking 'Why?' — typically five times — until the underlying, actionable cause is



WHEN TO USE

For any problem where the immediate cause is visible but the root cause (the one you can actually prevent) is not.

HOW TO APPLY — STEPS

1. State the problem clearly, specifically, and quantifiably
2. Ask Why it occurred and record the answer as a fact
3. Ask Why again about that answer
4. Repeat until you reach a cause you can act on at the system level
5. Verify: if fixing this root cause would prevent the problem recurring, you are done

PRACTITIONER TIP

! Each Why must be causally linked to the previous one. If you cannot draw a cause-effect chain, your analysis has gone wrong.

KEY OUTPUTS

- 5 Why's chain diagram or table
- Root cause statement
- Corrective action with verification date

HOW TO MEASURE SUCCESS

Problem recurrence rate at 30 and 90 days; time from detection to root cause identified

TOOL CATEGORY

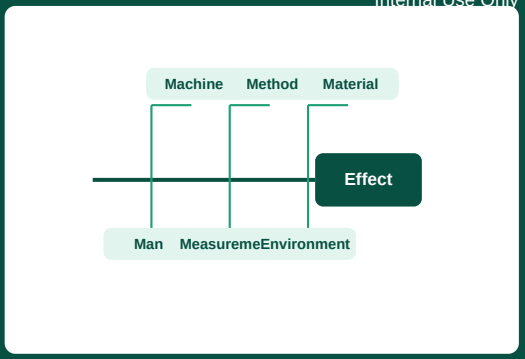
Quality & Process

QUALITY & PROCESS

Internal Use Only

7 Quality Tools

Seven graphical techniques for understanding and solving quality problems: Fishbone, Pareto, Run Chart, Histogram,



WHEN TO USE

Whenever quality data exists and patterns, causes, or variation need to be understood visually before deciding on action.

HOW TO APPLY — STEPS

1. Select the right tool for the specific question being asked
2. Collect clean, stratified data from the process
3. Build the chart using real data — not estimates
4. Interpret the patterns objectively
5. Use findings to direct root cause analysis and action

PRACTITIONER TIP

Start with a Pareto — it instantly shows where 80% of the problem lives. Don't analyse everything equally.

KEY OUTPUTS

- Pareto chart
- Fishbone diagram
- Control chart or run chart
- Histogram
- Check sheet

HOW TO MEASURE SUCCESS

Defect reduction in the prioritised category; time from problem detection to root cause identified

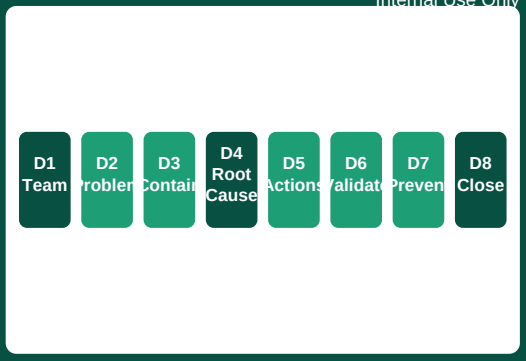
TOOL CATEGORY

Quality & Process

QUALITY & PROCESS

8D PSP

Resolve complex, recurring problems using an 8-step discipline that drives permanent corrective action and verified



WHEN TO USE

Customer complaints, repeat defects, safety incidents, or any issue requiring a rigorous, permanent fix with documentation.

HOW TO APPLY — STEPS

1. D1 — Form a cross-functional team with the right expertise
2. D2 — Describe the problem with data (5W2H)
3. D3 — Implement interim containment to protect the customer now
4. D4 — Identify and verify root cause and escape point
5. D5 — Select permanent corrective actions
6. D6 — Implement PCAs and validate effectiveness in production
7. D7 — Prevent recurrence — update all standards and control plans
8. D8 — Recognise the team and share the learning

PRACTITIONER TIP

! D3 containment must happen BEFORE D4 analysis begins. Protect the customer first, investigate second.

KEY OUTPUTS

- Completed 8D report
- Updated control plan and standards
- Customer notification if applicable

HOW TO MEASURE SUCCESS

Problem recurrence rate at 90 days; customer satisfaction score post-resolution

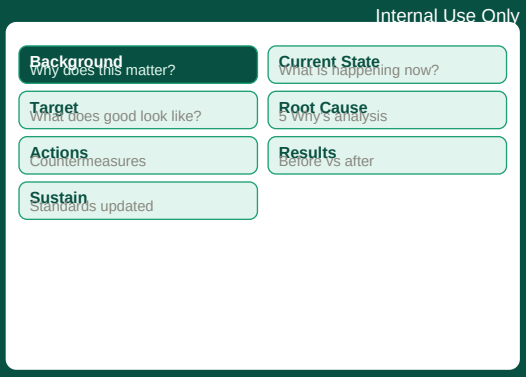
TOOL CATEGORY

Quality & Process

QUALITY & PROCESS

A3 PSP

Solve problems using a structured one-page A3 format covering background, current state, analysis,



WHEN TO USE

For medium-complexity problems requiring documented root cause analysis and a tested, verified countermeasure.

HOW TO APPLY — STEPS

1. Define background and business case for solving the problem
2. Describe current state with quantified data
3. Set the specific, measurable target condition
4. Analyse root causes using 5 Why's or fishbone diagram
5. Propose, test, and verify countermeasures
6. Standardise the solution and share learnings

PRACTITIONER TIP

The best A3s start with a pencil and a real problem observation, not a laptop and an assumption.

KEY OUTPUTS

- Completed A3 document
- Updated process standard
- Lessons learned summary

HOW TO MEASURE SUCCESS

Problem recurrence rate at 30 and 90 days; time from detection to root cause identification

TOOL CATEGORY

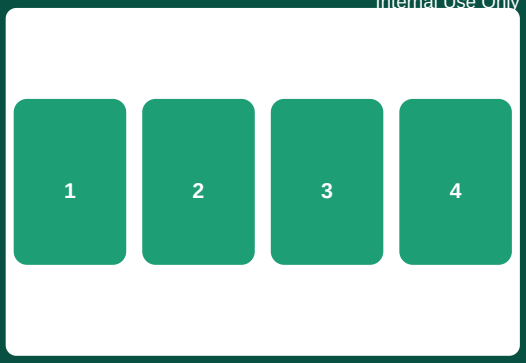
Quality & Process

QUALITY & PROCESS

Internal Use Only

Action CI Tracker

Track all continuous improvement ideas and actions on a single visible sheet — making CI activity transparent, owned,



WHEN TO USE

As a permanent feature of any team practising continuous improvement. Reviewed at every daily or weekly team meeting without exception.

HOW TO APPLY — STEPS

1. Capture every improvement idea the moment it is raised — no filtering at this stage
2. Assign an owner and a due date to every action before the meeting ends
3. Update status weekly: Open / In Progress / Complete / On Hold
4. Review overdue items at every meeting and agree action to close or escalate
5. Celebrate completions visibly — count ideas closed per month as a team KPI
6. Archive completed items monthly and review patterns for systemic improvements

PRACTITIONER TIP

! Post the CI tracker visibly in the team area. An idea captured and closed is worth ten ideas in a notebook. If it is not on the board, it does not exist.

KEY OUTPUTS

- Live CI action log with status and owner
- Monthly completion rate and idea velocity trend
- Escalation list for blocked or overdue actions
- CI culture metric: ideas per person per month

HOW TO MEASURE SUCCESS

Ideas generated per month; Ideas closed per month; Average days from idea to closure; % actions completed on time

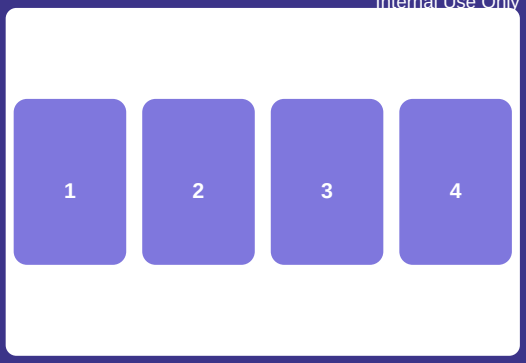
TOOL CATEGORY

Quality & Process

STRATEGY & PLANNING

Agile

Deliver work in short iterative cycles (sprints) with continuous feedback, enabling rapid adaptation to changing requirements.



WHEN TO USE

In development, innovation, or any environment where requirements evolve faster than traditional planning can handle.

HOW TO APPLY — STEPS

1. Build and prioritise a backlog of user stories
2. Plan a 2-week sprint — select top priority items
3. Execute with daily stand-ups to remove blockers
4. Sprint review: demonstrate working output to stakeholders
5. Retrospective: improve the way the team works itself

PRACTITIONER TIP

! 'Done' means fully done — not 90% complete. Never carry partially done work into the next sprint.

KEY OUTPUTS

- Prioritised sprint backlog
- Velocity chart
- Working deliverable each sprint
- Retrospective action log

HOW TO MEASURE SUCCESS

Velocity trend (story points per sprint); team satisfaction score; defects per sprint

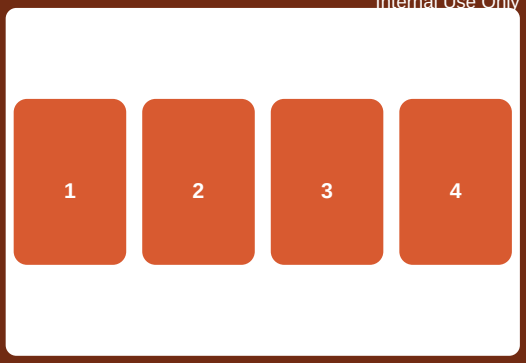
TOOL CATEGORY

Strategy & Planning

Internal Use Only

Alternative Solutions Evaluation

Systematically compare multiple solution options against agreed criteria to make an informed, transparent, and



WHEN TO USE

Before committing resources to a solution; when stakeholders disagree on the best approach and need an objective framework.

HOW TO APPLY — STEPS

1. Define 4–6 evaluation criteria relevant to the decision (cost, time, impact, risk, feasibility, fit)
2. Weight each criterion by relative importance to the decision context
3. Score each solution option against every criterion independently
4. Calculate weighted scores and rank options
5. Select the highest scorer or combine elements where appropriate
6. Document the full rationale for the decision and communicate to all stakeholders

PRACTITIONER TIP

! Always include the 'do nothing' option in every evaluation. It is usually the baseline against which all other options must be justified.

KEY OUTPUTS

- Options comparison matrix
- Weighted scoring table with rationale
- Decision record with justification
- Recommended solution with implementation plan

HOW TO MEASURE SUCCESS

Decision quality assessed at 6-month outcome review; stakeholder alignment score post-decision

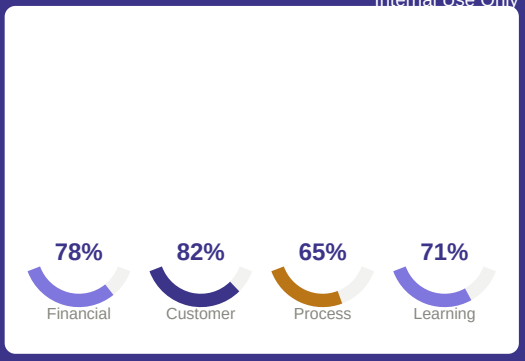
TOOL CATEGORY

People & CI

STRATEGY & PLANNING

Balanced Scorecard

Measure organisational performance across four perspectives: Financial, Customer, Internal Processes, and Learning &



WHEN TO USE

When decisions are driven by financial metrics alone; when strategy lacks measurable targets across all dimensions.

HOW TO APPLY — STEPS

1. Define strategic objectives per perspective
2. Identify KPIs for each objective
3. Set targets and link to initiatives
4. Review monthly or quarterly
5. Communicate results to all levels

PRACTITIONER TIP

Link KPIs to real decisions — avoid vanity metrics that no one acts on. Every KPI must have an owner.

KEY OUTPUTS

- One-page scorecard
- KPI dashboard
- Strategy map

HOW TO MEASURE SUCCESS

KPI achievement rate per perspective; number of strategic initiatives on track

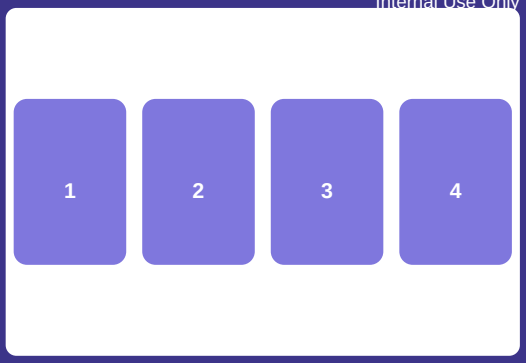
TOOL CATEGORY

Strategy & Planning

STRATEGY & PLANNING

Benchmarking

Compare performance, processes, or practices against industry leaders to identify improvement opportunities and set



WHEN TO USE

When unsure how competitive your performance is; before setting stretch targets or making investment decisions.

HOW TO APPLY — STEPS

1. Define what to benchmark (process, metric, practice)
2. Identify best-in-class comparators
3. Collect data through research, site visits, or industry reports
4. Analyse gaps and root causes
5. Adapt and implement best practices contextually

PRACTITIONER TIP

! Focus on understanding WHY the leader performs better, not just copying WHAT they do. Context always matters.

KEY OUTPUTS

- Benchmark comparison report
- Gap analysis
- Improvement roadmap

HOW TO MEASURE SUCCESS

Gap closed vs benchmark over 12 months; performance percentile improvement

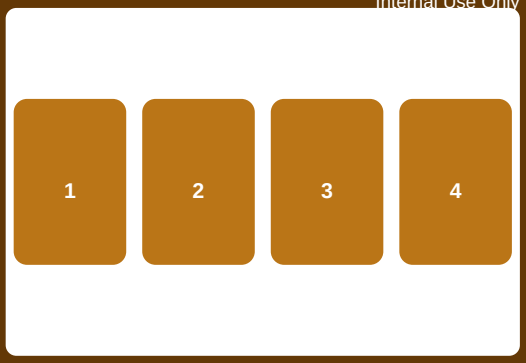
TOOL CATEGORY

Strategy & Planning

OPERATIONS

Bottleneck Walk

Physically walk a process to identify the specific constraining step that limits overall system throughput.



WHEN TO USE

When total output is below target but individual steps appear busy; when lead time is unexpectedly or consistently long.

HOW TO APPLY — STEPS

1. Walk the process physically from start to finish
2. Observe and measure queue lengths at each step — the longest queue is upstream of the bottleneck
3. Measure actual cycle times at each process step
4. Confirm the step with the lowest effective capacity is the bottleneck
5. Focus ALL available improvement resources on that single constraint

PRACTITIONER TIP

! The bottleneck is almost never where management thinks it is. Always walk before drawing any conclusions.

KEY OUTPUTS

- Bottleneck identification with evidence
- Cycle time comparison chart across all steps
- Capacity model showing the constraint

HOW TO MEASURE SUCCESS

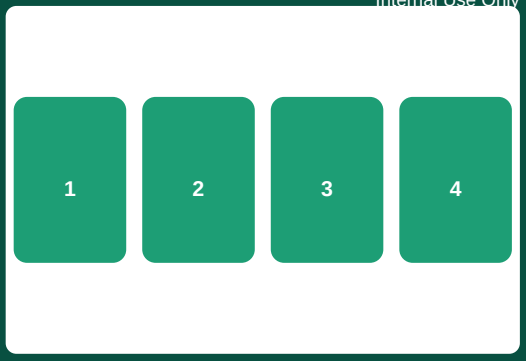
System throughput increase after bottleneck action; queue length reduction at the constraint

TOOL CATEGORY
Operations

QUALITY & PROCESS

Box & Whisker Plot

Visualise the distribution of a dataset — median, quartiles, spread, and outliers — in a compact graphical format for



WHEN TO USE

Comparing distributions across groups — shifts, machines, operators, suppliers, time periods, or product lines.

HOW TO APPLY — STEPS

1. Collect data for each comparison group
2. Calculate median, Q1, Q3, and whisker bounds
3. Plot and compare distributions visually side by side
4. Identify which groups have the widest spread or most outliers
5. Investigate root causes of differences between groups

PRACTITIONER TIP

! A wide box means high variation — usually more important to address than a shifted mean. Reduce spread first.

KEY OUTPUTS

- Box plot chart
- Summary statistics table
- Comparison insights and hypotheses

HOW TO MEASURE SUCCESS

Interquartile range reduction; outlier frequency reduction; between-group variation reduction

TOOL CATEGORY

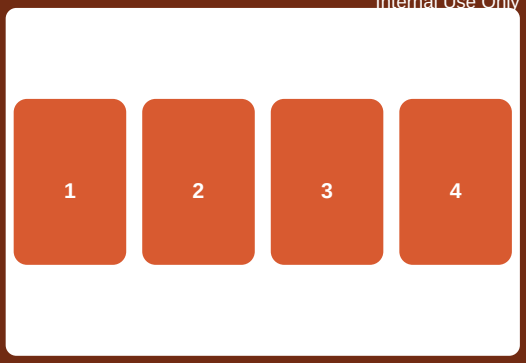
Quality & Process

PEOPLE & CI

Internal Use Only

Brainstorming

Generate a large quantity of creative ideas in a structured, judgment-free group session to expand the solution or



WHEN TO USE

When looking for creative solutions; at the start of problem solving or improvement planning before narrowing options.

HOW TO APPLY — STEPS

1. Define the question or challenge clearly and specifically in advance — before people arrive
2. Set and enforce the rule: no evaluation, no criticism, no 'yes, but' during idea generation
3. Generate ideas in silence individually for 3–5 minutes before sharing with the group
4. Capture all ideas visibly; cluster related ideas into themes
5. Vote or use criteria to shortlist the top candidates for further development

PRACTITIONER TIP

! Quantity before quality every time. The best ideas in a session almost always emerge after you have already generated 30 others.

KEY OUTPUTS

- Full idea list (no filtering during generation)
- Clustered themes and patterns
- Shortlist of top ideas for development

HOW TO MEASURE SUCCESS

Ideas generated per session; % of ideas advanced to action; quality of selected ideas rated at 90-day outcome review

TOOL CATEGORY

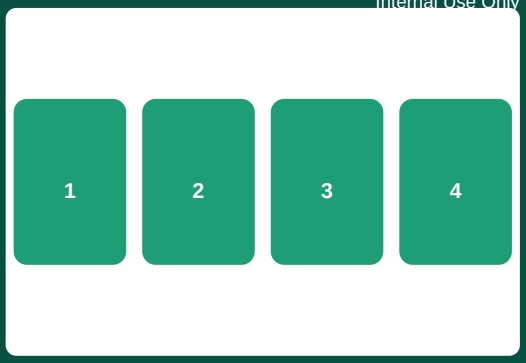
People & CI

QUALITY & PROCESS

Internal Use Only

CI Activity Sheet

Track individual or team continuous improvement ideas and actions on a single visible sheet — making all CI activity



WHEN TO USE

As a permanent feature of any team practising continuous improvement — reviewed every week without exception.

HOW TO APPLY — STEPS

1. Define standard columns: idea description, owner, target date, and status
2. Post visibly at the team's daily management board
3. Review and update at every weekly team meeting
4. Celebrate completed improvements publicly
5. Escalate items blocked for more than two weeks to the line manager

PRACTITIONER TIP

! Quantity of ideas matters far less than quality of completions. A wall covered in sticky notes with nothing implemented is not CI.

KEY OUTPUTS

- CI activity log visible to the whole team
- Completion trend chart
- List of items needing escalation

HOW TO MEASURE SUCCESS

Ideas completed per month; average days from idea to closure; completion rate %

TOOL CATEGORY

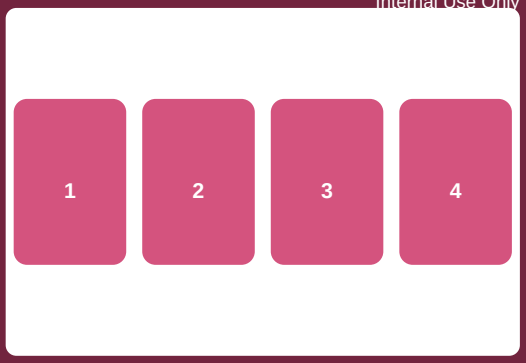
Quality & Process

COMMERCIAL

Internal Use Only

CRM Process Review

Audit and improve how CRM processes and tools are being used to ensure data quality, pipeline accuracy, and



WHEN TO USE

When forecast accuracy is consistently poor; when CRM data is incomplete or distrusted by leadership; after implementing a new CRM system.

HOW TO APPLY — STEPS

1. Define what 'good' CRM usage looks like: required fields, stage definitions, update frequency, activity logging
2. Audit the current state — data completeness, accuracy, and consistency across the team
3. Identify specific usage gaps and the process failures causing them
4. Train and coach on correct usage with worked examples from your own data
5. Measure data quality monthly and report as a team KPI

PRACTITIONER TIP

! CRM quality is a leadership discipline issue before it is a technology issue. If leaders do not demand quality data, teams will not supply it.

KEY OUTPUTS

- CRM data quality audit report
- Process improvement plan with accountabilities
- Training and coaching materials
- Monthly data quality dashboard

HOW TO MEASURE SUCCESS

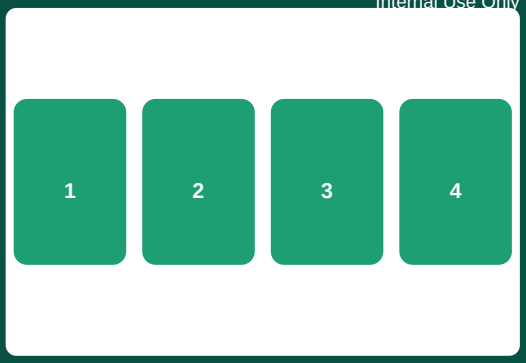
CRM data completeness %; forecast accuracy improvement vs baseline; pipeline coverage ratio

TOOL CATEGORY
Commercial

Internal Use Only

Chalk Circle

Develop deep process observation skills by standing in one location for an extended period and systematically recording



WHEN TO USE

Before designing improvements; when assumptions about how a process actually works need to be tested against reality.

HOW TO APPLY — STEPS

1. Select a critical observation point in the process
2. Stand or mark the spot — observe for 30–60 minutes without interrupting
3. Record everything: activities, problems, waste, variation, unsafe conditions
4. Do not intervene during the observation period
5. Debrief findings with the improvement team immediately after

PRACTITIONER TIP

! Bring a notebook, not a laptop. The act of handwriting slows you down and forces you to notice what you would otherwise filter out.

KEY OUTPUTS

- Structured observation log
- Problem and waste priority list
- Insights that reshape the improvement plan

HOW TO MEASURE SUCCESS

Number of problems identified per observation hour; insights that change the improvement direction

TOOL CATEGORY

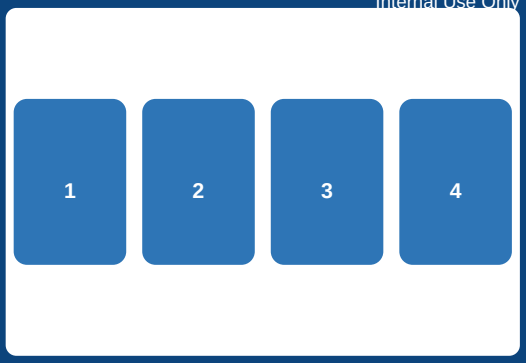
Quality & Process

MANAGEMENT SYSTEMS

Internal Use Only

Communication Standard

Define how information flows between roles, teams, and levels — specifying what is communicated, by whom, in what format,



WHEN TO USE

When people repeatedly say 'I didn't know' or 'nobody told me'; when the same information is shared in multiple inconsistent ways.

HOW TO APPLY — STEPS

1. Map current communication flows and identify all gaps and duplication
2. Design the standard: who communicates what, to whom, when, and in what format
3. Document and display the communication matrix at team level
4. Train all relevant roles on the new standard
5. Audit compliance monthly and refine based on feedback

PRACTITIONER TIP

! Less is more. A communication standard that tries to include everything becomes too complex to follow and ends up used by no one.

KEY OUTPUTS

- Communication matrix
- Standard message templates
- Monthly compliance audit

HOW TO MEASURE SUCCESS

Information receipt confirmation rate; communication-related errors or 'I didn't know' incidents per month

TOOL CATEGORY

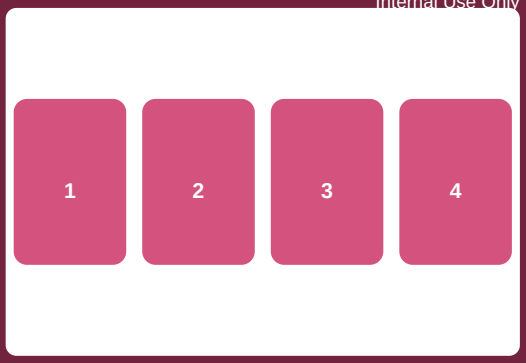
Management Systems

COMMERCIAL

Internal Use Only

Competitive Analysis

Systematically research and compare competitors on product, price, positioning, channels, and strategy to identify threats,



WHEN TO USE

Before launching a new product or market; when win rates against specific competitors are declining; during annual strategic planning.

HOW TO APPLY — STEPS

1. Define the competitive set clearly — direct and indirect competitors relevant to each market segment
2. Research each competitor across agreed dimensions: features, pricing, positioning, channels, messaging, financials
3. Identify specifically where you are stronger and where you are weaker than each competitor
4. Define a strategic response for each material gap or vulnerability identified
5. Update the analysis at least quarterly — competitive positions shift faster than annual reviews can track

PRACTITIONER TIP

! Base competitive analysis on observable evidence — not on anecdotes from the sales team about why they lost deals last month.

KEY OUTPUTS

- Competitive comparison matrix
- Market positioning map
- Threat and opportunity assessment
- Strategic response plan with prioritised actions

HOW TO MEASURE SUCCESS

Win rate trend vs specific named competitors; market share position; positioning differentiation score from customer survey

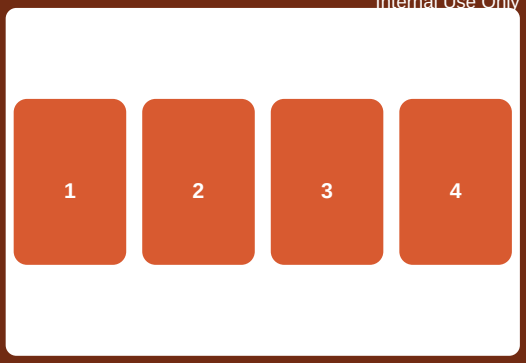
TOOL CATEGORY
Commercial

PEOPLE & CI

Internal Use Only

Customer Feedback

Systematically collect, analyse, and act on customer feedback to drive measurable improvement and build customer loyalty.



WHEN TO USE

Continuously as part of normal operations — not only when there is a problem or a management review approaching.

HOW TO APPLY — STEPS

1. Define all feedback channels: survey, NPS, complaints, direct conversation, social, support tickets
2. Collect consistently and embed in the normal customer journey — not as an afterthought
3. Analyse themes monthly and identify the top three issues by frequency and impact
4. Present issues and owners at every management review
5. Track improvement and close the loop with customers who gave feedback — tell them what changed

PRACTITIONER TIP

! Collecting feedback without acting on it is actively worse than not collecting it. It signals to customers that you do not care.

KEY OUTPUTS

- Customer feedback dashboard with trend
- Theme analysis report
- Action log with owners and due dates
- Closed-loop customer communication

HOW TO MEASURE SUCCESS

NPS trend; complaint resolution time in days; % of feedback themes actioned within 90 days

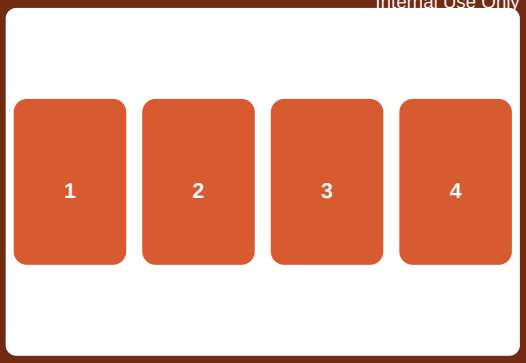
TOOL CATEGORY
People & CI

PEOPLE & CI

Internal Use Only

Customer Gemba

Visit the customer's actual environment to directly observe how they use your product or service — generating insight



WHEN TO USE

Before redesigning a product or service; when satisfaction scores are falling; when customer needs are assumed rather than observed.

HOW TO APPLY — STEPS

1. Define specific observation objectives before the visit
2. Go to the customer's location — do not invite them to yours
3. Observe and listen without defending, selling, or explaining
4. Ask open questions: 'Can you show me exactly how you use this?'
5. Debrief the team immediately after and integrate insights into improvement plans

PRACTITIONER TIP

! The most important word in 'Customer Gemba' is Gemba — the real place. Insights come from observation, not surveys or meetings.

KEY OUTPUTS

- Structured observation notes
- Customer insight summary with direct quotes
- Prioritised improvement opportunities ranked by customer impact

HOW TO MEASURE SUCCESS

Insights generated per visit; improvement actions taken per quarter; NPS or CSAT before and after acting on insights

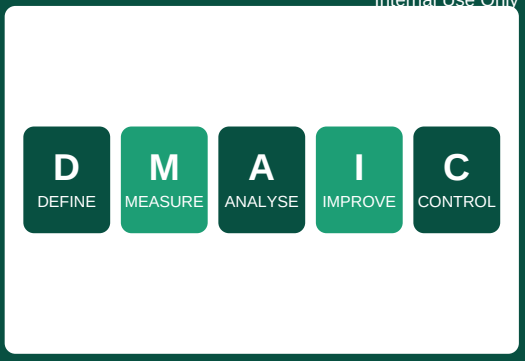
TOOL CATEGORY
People & CI

QUALITY & PROCESS

Internal Use Only

DMAIC

A rigorous data-driven improvement method — Define, Measure, Analyse, Improve, Control — used in Six Sigma



WHEN TO USE

For complex problems with significant financial or quality impact where data is available or can be collected.

HOW TO APPLY — STEPS

1. Define — problem scope, team, project charter, and customer requirements
2. Measure — baseline process capability and collect reliable data
3. Analyse — identify root causes using statistical methods
4. Improve — generate, test, pilot, and select optimal solutions
5. Control — sustain gains with monitoring plans, control charts, and updated standards

PRACTITIONER TIP

! Never skip the Measure phase. Jumping to solutions without baseline data is the most common and costly DMAIC mistake.

KEY OUTPUTS

- Signed project charter
- Process capability report (Cp, Cpk)
- Root cause analysis
- Control plan

HOW TO MEASURE SUCCESS

Sigma level improvement; financial benefit delivered ($); defect reduction percentage

TOOL CATEGORY

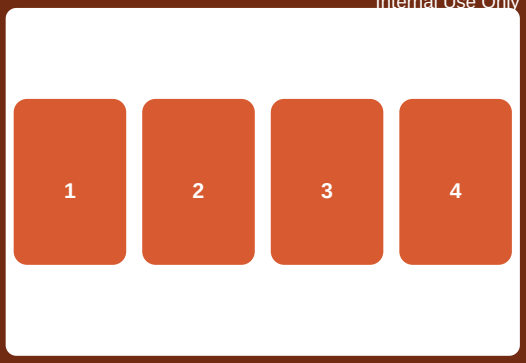
Quality & Process

PEOPLE & CI

Internal Use Only

Design Thinking

A human-centred innovation process using empathy, ideation, and rapid prototyping to generate creative solutions to



WHEN TO USE

For customer experience challenges, new service design, product innovation, or any problem where the solution space is genuinely unknown.

HOW TO APPLY — STEPS

1. Empathise — deeply understand the user through observation and open interviews
2. Define — synthesise findings into a clear, specific 'How Might We' problem statement
3. Ideate — generate a wide range of ideas without judgment or filtering
4. Prototype — build rough, fast, cheap representations of the top two or three ideas
5. Test — get real user feedback on prototypes and iterate based on what you learn

PRACTITIONER TIP

! The prototype does not need to work — it needs to teach. The goal is fast learning, not a finished solution on the first attempt.

KEY OUTPUTS

- Empathy map and user journey
- Problem statement (How Might We)
- Prototype or mock-up
- User test insights and iteration plan

HOW TO MEASURE SUCCESS

Number of user tests completed per idea; insights per test session; adoption rate of final solution at 6 months

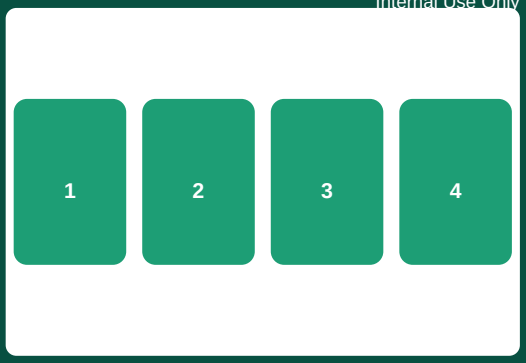
TOOL CATEGORY

People & CI

Internal Use Only

Design of Experiments

Systematically test combinations of input variables to find which factors most influence an output — and identify optimal



WHEN TO USE

When multiple factors may interact; when optimising a process or formulation; when one-factor-at-a-time testing is too slow or costly.

HOW TO APPLY — STEPS

1. Define the response variable (output) to be optimised
2. Identify input factors and levels to test
3. Choose experimental design (full factorial, fractional, Taguchi, RSM)
4. Run experiments in randomised order to avoid confounding
5. Analyse main effects and interactions
6. Confirm optimal settings with a separate validation run

PRACTITIONER TIP

! Randomise your run order without exception. Systematic patterns create confounding that invalidates all results.

KEY OUTPUTS

- Experimental design matrix
- Main effects and interaction plots
- Optimal factor settings
- Validation run confirmation

HOW TO MEASURE SUCCESS

% improvement in response variable; experimental efficiency vs one-factor-at-a-time

TOOL CATEGORY

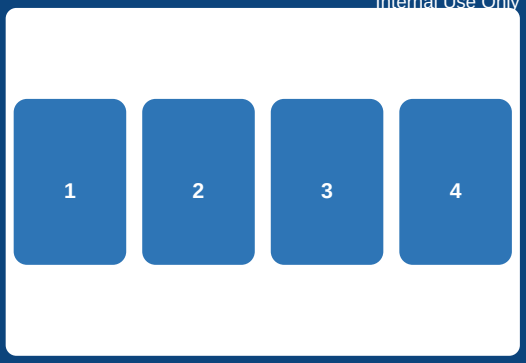
Quality & Process

MANAGEMENT SYSTEMS

Internal Use Only

Deviation of Standard Procedure

Manage and investigate deviations from documented standards to understand why they occur and permanently



WHEN TO USE

Whenever a team member does not follow the defined procedure — whether by mistake, necessity, or choice.

HOW TO APPLY — STEPS

1. Detect and document the deviation immediately: what, where, when, who, and what actually happened
2. Determine immediate impact on safety, quality, or delivery, and contain if necessary
3. Investigate root cause: Was the standard unclear? Training insufficient? Standard incorrect? Environment unsuitable?
4. Take corrective action at root cause level — retrain, update the standard, or redesign the process
5. Verify within 30 days that the deviation has not recurred

PRACTITIONER TIP

! Most deviations reveal a problem with the standard or the system, not the person. Approach every deviation with curiosity, not blame.

KEY OUTPUTS

- Deviation record with investigation
- Root cause and corrective action documentation
- Updated standard or training record where applicable

HOW TO MEASURE SUCCESS

Deviation frequency trend; % with root cause investigation completed; recurrence rate at 30 days

TOOL CATEGORY

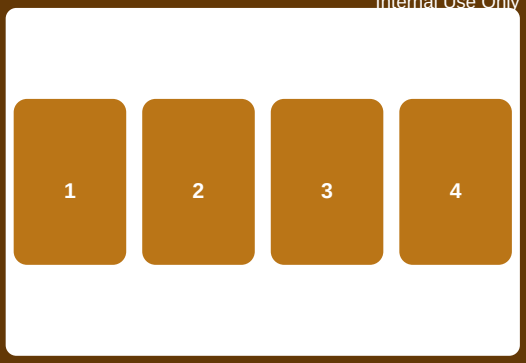
Management Systems

OPERATIONS

Internal Use Only

Early Equipment Management (EEM)

Involve operations and maintenance teams in equipment specification, design, and commissioning to achieve fast



WHEN TO USE

When procuring new equipment or designing new production lines — before the specification is issued to vendors.

HOW TO APPLY — STEPS

1. Involve production and maintenance engineers at the specification stage — before purchase
2. Capture and integrate lessons learned from existing similar equipment
3. Review equipment designs formally for maintainability and operability
4. Plan the vertical start-up with detailed commissioning milestones
5. Target achieving designed performance level within the first month of operation

PRACTITIONER TIP

! The team that operates and maintains the equipment must help specify it. Design engineers working alone consistently get it wrong.

KEY OUTPUTS

- Equipment specification incorporating operational requirements
- Maintenance Prevention (MP) information database
- Vertical start-up plan
- Commissioning performance report

HOW TO MEASURE SUCCESS

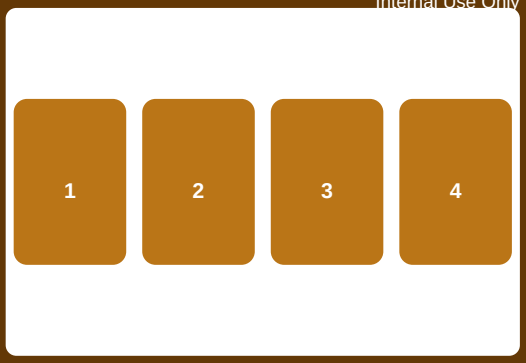
Days to achieve target OEE after commissioning; early-life failures vs benchmark; ramp-up cost vs plan

TOOL CATEGORY
Operations

OPERATIONS

Early Product Development (EPD)

Apply lean and concurrent engineering principles to the development phase to shorten cycles, reduce engineering



WHEN TO USE

When product launches are consistently late, over budget, or followed by high defect rates in early production.

HOW TO APPLY — STEPS

1. Define customer value explicitly before designing any technical solution
2. Use set-based concurrent engineering — develop multiple options in parallel
3. Apply obeya (visual project rooms) to make progress and decisions visible
4. Conduct regular technical reviews with the chief engineer or equivalent
5. Apply FMEA and Design for X at the design stage — not after tooling is committed

PRACTITIONER TIP

! The cost of changing a design doubles at every gate. Fix problems on paper and simulation, not in tooling.

KEY OUTPUTS

- Obeya room with live project status
- Set-based design options comparison
- Engineering FMEA
- Knowledge-based decisions log

HOW TO MEASURE SUCCESS

Engineering change orders post-launch; time-to-market vs plan; first-time quality in production launch month

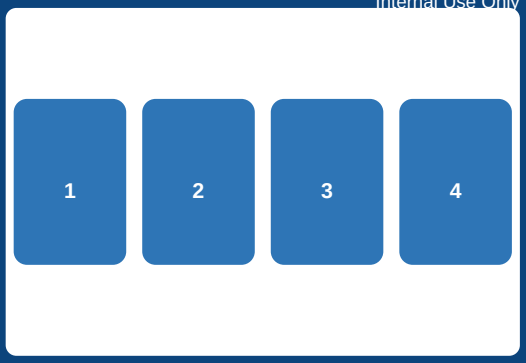
TOOL CATEGORY
Operations

MANAGEMENT SYSTEMS

Internal Use Only

Effective Meeting Guide

Provide a simple framework to make every meeting productive — with clear purpose, active participation, and documented



WHEN TO USE

For any recurring or project meeting — to raise the overall meeting culture of a team or department.

HOW TO APPLY — STEPS

1. Confirm objective before the meeting begins
2. Start and end on time — without exception
3. One person speaks at a time; facilitator manages flow
4. Capture decisions and actions in real time during the meeting
5. Circulate minutes and action log within 24 hours

PRACTITIONER TIP

! Cancel any meeting where the objective can be achieved with a clear email or message. Protect people's time.

KEY OUTPUTS

- Meeting agenda
- Action and decision log
- Follow-up tracking

HOW TO MEASURE SUCCESS

Action completion rate from meetings %; average meeting duration vs plan

TOOL CATEGORY

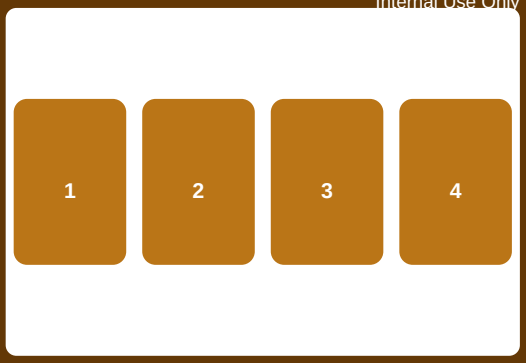
Management Systems

OPERATIONS

Internal Use Only

Equipment Performance Factor

Measure and track the actual performance of individual equipment against its designed output rate to identify and



WHEN TO USE

When machine output is inconsistent or below standard; before a capital investment decision on replacement or upgrade.

HOW TO APPLY — STEPS

1. Define the rated (nameplate) capacity for the machine under normal conditions
2. Measure actual output over at least two representative production weeks
3. Calculate performance factor = actual output / rated output x 100%
4. Investigate root causes of the performance gap by shift and product
5. Implement and verify improvement actions systematically

PRACTITIONER TIP

! Measure over at least two full weeks to capture shift-to-shift variation and recurring faults across the full cycle.

KEY OUTPUTS

- Performance factor trend chart by machine
- Loss analysis by category
- Targeted improvement action plan

HOW TO MEASURE SUCCESS

Performance factor % increase; variability reduction between shifts; comparison to rated specification

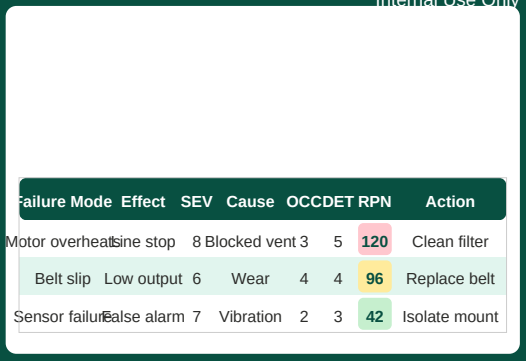
TOOL CATEGORY
Operations

QUALITY & PROCESS

Internal Use Only

FMEA

Proactively identify potential failure modes, their effects, causes, and controls before they occur — then prioritise risk



Failure Mode	Effect	SEV	Cause	OCC	DET	RPN	Action
Motor overheats	Line stop	8	Blocked vent	3	5	120	Clean filter
Belt slip	Low output	6	Wear	4	4	96	Replace belt
Sensor failure	False alarm	7	Vibration	2	3	42	Isolate mount

WHEN TO USE

Before launching a new process or product; when designing controls for high-risk steps; after a serious incident.

HOW TO APPLY — STEPS

1. List all process steps or product functions
2. Identify potential failure modes for each step
3. Assess Severity, Occurrence, and Detection each on a 1–10 scale
4. Calculate RPN = Severity × Occurrence × Detection
5. Prioritise and action the highest RPN items first
6. Re-score after actions to confirm RPN has reduced

PRACTITIONER TIP

Prioritise HIGH Severity items regardless of RPN. Some failures are unacceptable no matter how rarely they occur.

KEY OUTPUTS

- FMEA table with RPN scores
- Risk priority ranking
- Action plan with verified revised RPNs

HOW TO MEASURE SUCCESS

Average RPN reduction %; number of critical failure modes with effective mitigation

TOOL CATEGORY

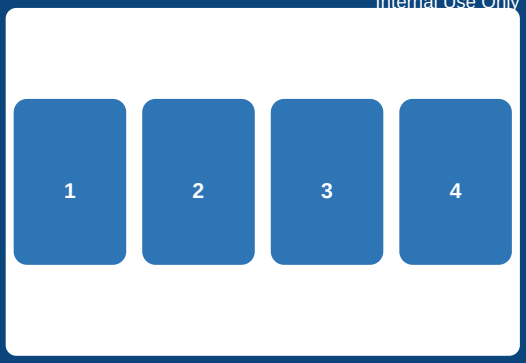
Quality & Process

MANAGEMENT SYSTEMS

Internal Use Only

Feedback Rules

Define how feedback is given and received within a team to create psychological safety and ensure feedback actually



WHEN TO USE

When teams avoid honest conversation; when feedback is ad hoc, absent, or consistently perceived as personal attack.

HOW TO APPLY — STEPS

1. Define the team's feedback principles — timely, specific, behaviour-focused, two-way, forward-looking
2. Train all team members on both giving and receiving feedback effectively
3. Leadership must model feedback from the top down and visibly receive it
4. Create regular structured feedback moments: 1:1s, team retrospectives, project reviews
5. Measure whether feedback consistently leads to visible and acknowledged change

PRACTITIONER TIP

! Feedback about a person's character is never constructive. Feedback about a specific, observable behaviour in a specific situation always is.

KEY OUTPUTS

- Team feedback charter
- Training workshop materials
- 1:1 conversation template

HOW TO MEASURE SUCCESS

Feedback frequency per person per month; team psychological safety score; visible behaviour changes post-feedback

TOOL CATEGORY

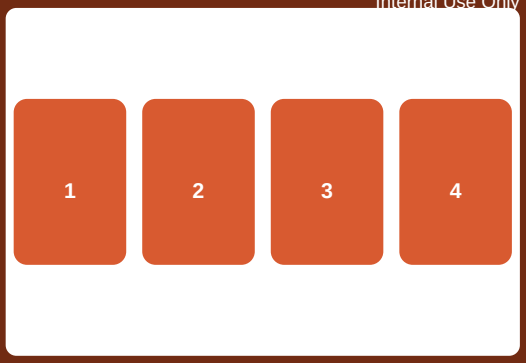
Management Systems

PEOPLE & CI

Internal Use Only

GO & See

Leaders leave the office and visit the actual workplace to observe firsthand — replacing management by reports with



WHEN TO USE

Daily for leaders at all levels. GO & See is a habit and discipline, not an occasional event or inspection.

HOW TO APPLY — STEPS

1. Schedule regular GO & See visits with a defined frequency — don't leave it to when you have time
2. Use a structured observation checklist to maintain consistency
3. Observe before intervening — ask questions first, give opinions second
4. Provide coaching and support to team leaders based on what you see
5. Follow up on every identified issue within 24–48 hours to demonstrate credibility

PRACTITIONER TIP

! GO & See is coaching, not inspecting. If the team hides problems when you arrive, your management approach — not theirs — needs to change.

KEY OUTPUTS

- GO & See observation log
- Coaching and development notes
- Action items for management follow-up

HOW TO MEASURE SUCCESS

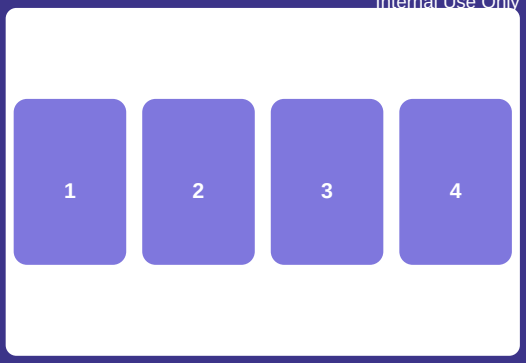
GO & See visit frequency per week; issues identified per visit; team engagement score trend

TOOL CATEGORY
People & CI

STRATEGY & PLANNING

Gantt Chart

Visualise project tasks, dependencies, durations, and timelines on a horizontal bar chart to plan and track progress.



WHEN TO USE

Planning any multi-task project where sequence, timing, and dependencies need to be visible.

HOW TO APPLY — STEPS

1. List all tasks required to complete the project
2. Estimate duration for each task
3. Identify dependencies between tasks
4. Plot on a timeline showing critical path
5. Update weekly as actual progress diverges from plan

PRACTITIONER TIP

Show the critical path prominently. All other detail is useful but the critical path is what matters for delivery.

KEY OUTPUTS

- Gantt chart with critical path
- Resource loading view
- Weekly progress report

HOW TO MEASURE SUCCESS

Tasks completed on time %; critical path slippage in days

TOOL CATEGORY

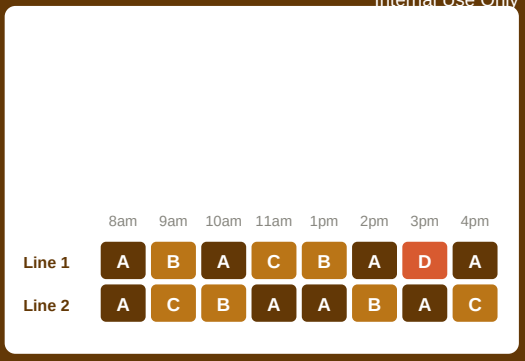
Strategy & Planning

OPERATIONS

Internal Use Only

Heijunka

Level production volume and mix across a planning period to smooth demand on upstream processes, workforce, and



WHEN TO USE

When production scheduling creates peaks and valleys that stress resources and cause excessive overtime and idle time alternately.

HOW TO APPLY — STEPS

1. Calculate average daily demand per product family over 4–8 weeks
2. Create a heijunka box with defined time slots
3. Load orders evenly across the planning period by volume and mix
4. Communicate the levelled schedule to production and supply
5. Measure schedule adherence and work to reduce variation

PRACTITIONER TIP

! You cannot level production while sales promises custom delivery windows to every customer. Align commercial and operations first.

KEY OUTPUTS

- Heijunka scheduling box
- Levelled production schedule
- Volume smoothing trend chart

HOW TO MEASURE SUCCESS

Production levelling index; schedule adherence rate; WIP inventory reduction %

TOOL CATEGORY

Operations

STRATEGY & PLANNING

Hoshin Kanri

Align company strategy to daily operations by cascading goals from leadership to frontline teams through a structured



WHEN TO USE

Annual strategic planning cycles; when teams work in silos with no visible link to company objectives.

HOW TO APPLY — STEPS

1. Define 3–5 breakthrough objectives at executive level
2. Cascade goals to departmental X-matrices
3. Align team-level action plans to department targets
4. Review monthly — adjust quarterly
5. Conduct annual reflection and reset

PRACTITIONER TIP

Keep to 3–5 priorities. More than 5 dilutes focus and guarantees failure in execution.

KEY OUTPUTS

- X-Matrix (strategy on one page)
- Departmental action plans
- Monthly review dashboards

HOW TO MEASURE SUCCESS

% of KPIs on track; % of action plans completed on time

TOOL CATEGORY

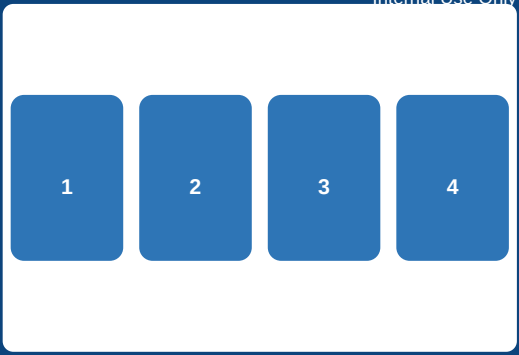
Strategy & Planning

MANAGEMENT SYSTEMS

Internal Use Only

Introduction of a Standard

Follow a structured process when introducing new standards so they are genuinely understood, accepted, and followed —



WHEN TO USE

Whenever a new process, procedure, policy, or work instruction is being created and deployed.

HOW TO APPLY — STEPS

1. Involve the people who will use the standard in writing it — do not write it for them
2. Pilot the standard with a small group and capture improvements
3. Train all affected users formally with demonstration, not just document distribution
4. Display the standard prominently at the point of use
5. Verify application in practice within 30 days and capture deviations

KEY OUTPUTS

- New standard document with version number
- Pilot test results and improvements
- Training completion record
- 30-day adherence verification audit

HOW TO MEASURE SUCCESS

Standard adherence rate at 30 and 90 days; deviations or errors in the first month of deployment

PRACTITIONER TIP

! A standard issued by email and expected to be self-taught will not be followed. The medium is the message when it comes to standards.

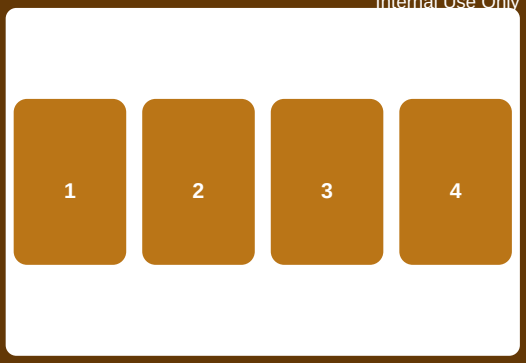
TOOL CATEGORY

Management Systems

OPERATIONS

JIT (Just-In-Time)

Produce or deliver exactly what is needed, when it is needed, in the exact quantity required — eliminating overproduction



WHEN TO USE

When inventory levels are high, storage costs are significant, or lead times need to be dramatically shortened.

HOW TO APPLY — STEPS

1. Map current material flow and identify all inventory buffers
2. Define takt time based on actual customer demand rate
3. Size production quantities to match demand signal, not forecast
4. Implement pull triggers using kanban or actual demand signals
5. Stabilise supply chain reliability before reducing safety stock

PRACTITIONER TIP

! JIT fails without reliable supply. Fix supplier on-time delivery before you reduce safety stock — sequence matters.

KEY OUTPUTS

- Material flow map
- Pull system design document
- Inventory reduction phasing plan

HOW TO MEASURE SUCCESS

Inventory turns per year; days on hand reduction; overproduction incidents per month

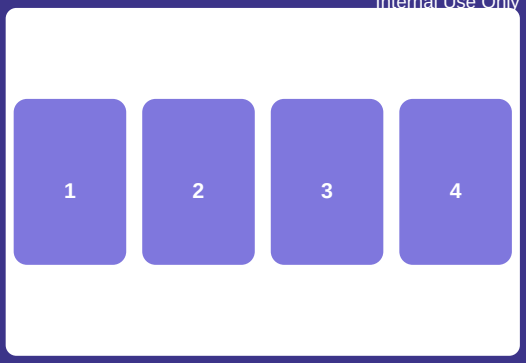
TOOL CATEGORY
Operations

STRATEGY & PLANNING

Internal Use Only

KPI Check

Systematically review key performance indicators to confirm whether performance is on track and to trigger immediate



WHEN TO USE

Daily, weekly, and monthly at every level of the organisation as part of routine management.

HOW TO APPLY — STEPS

1. Define KPIs with clear targets and measurement methodology
2. Collect and display data consistently on visual boards
3. Review at daily and weekly meetings using RAG status
4. Investigate RED indicators within 24 hours
5. Track improvement actions to closure

PRACTITIONER TIP

A KPI with no owner is a decoration. Assign a named person to every metric — not a department, a person.

KEY OUTPUTS

- KPI dashboard with RAG status
- Trend charts
- Action log for off-target indicators

HOW TO MEASURE SUCCESS

% KPIs within target; response time to RED indicators in hours

TOOL CATEGORY

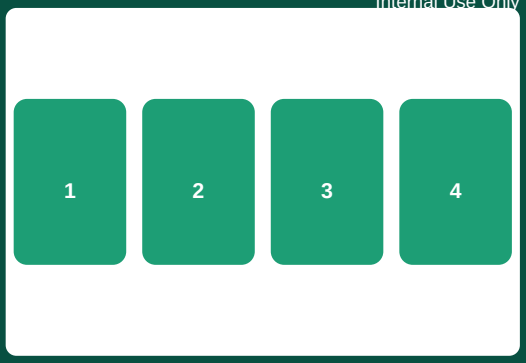
Strategy & Planning

QUALITY & PROCESS

Internal Use Only

Kaizen Event Report

Document a structured 3–5 day focused improvement event from objectives set on Day 1 through to results measured on



WHEN TO USE

During and immediately after every Kaizen event. The report is a living document — started on Day 1 and completed before the team disperses on Day 5.

HOW TO APPLY — STEPS

1. Day 1: Document event objectives, team, scope, and current state baseline data
2. Day 2: Record gemba observations, root cause analysis findings, and waste identified
3. Day 3–4: Capture every improvement implemented during the event with photos
4. Day 4–5: Measure before vs after results using the same metrics as the baseline
5. Confirm which standards and work instructions have been updated
6. Agree sustain audit schedule — 30, 60, and 90-day check dates with named auditors
7. Document open actions with owners and due dates (maximum 10 items, max 30 days)
8. Record lessons learned and propose the next Kaizen event area

PRACTITIONER TIP

! If results are not measured on Day 4, they will never be measured. Never leave the event without an agreed sustain audit date in the calendar and a named auditor.

KEY OUTPUTS

- Completed Kaizen event report with before/after data
- Photo evidence of changes made during the event
- Updated standards and work instructions list
- Sustain audit schedule (30/60/90 days) with owners
- Open actions log with owners and due dates

HOW TO MEASURE SUCCESS

% of targets achieved vs Day 1 goals; Sustained at 30-day audit Y/N; Open actions closed within 30 days %; Sustain audit compliance rate

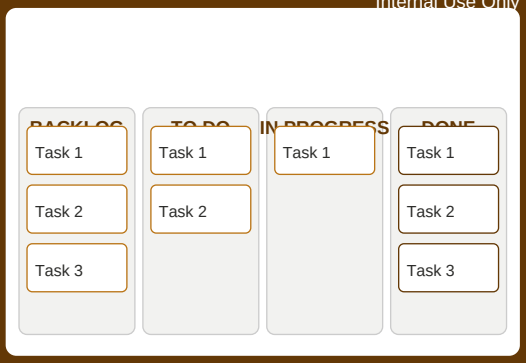
TOOL CATEGORY

Quality & Process

OPERATIONS

Kanban

Control inventory and workflow using visual signals — cards, bins, or board columns — that trigger replenishment only



WHEN TO USE

To replace push scheduling with a demand-driven pull system; to make work in progress visible and strictly limited.

HOW TO APPLY — STEPS

1. Analyse consumption rate and replenishment lead time for each item
2. Calculate the required kanban quantity for each part or task
3. Design the trigger mechanism — physical card, empty bin, or board column
4. Train all users thoroughly before launch
5. Review and adjust kanban quantities regularly as demand patterns change

PRACTITIONER TIP

! Start with too many kanbans and reduce them gradually over weeks. Starting with too few causes stoppages on day one.

KEY OUTPUTS

- Kanban design specification
- Physical or digital kanban system
- WIP reduction trend chart

HOW TO MEASURE SUCCESS

WIP reduction %; stockout frequency per week; inventory value reduction

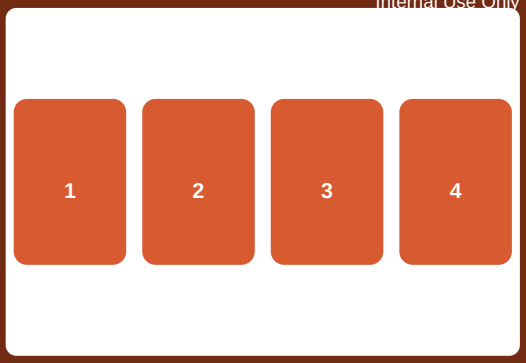
TOOL CATEGORY
Operations

PEOPLE & CI

Internal Use Only

Leading a Workshop

Design and facilitate a group working session that achieves a defined objective through structured, inclusive participation.



WHEN TO USE

For problem solving, strategic decision making, improvement planning, or team alignment on complex topics.

HOW TO APPLY — STEPS

1. Define the workshop objective and specific desired outputs before designing the agenda
2. Design the agenda with timed activities matched to the energy curve of the group
3. Prepare all materials, tools, and room layout in advance
4. Open with a clear statement of purpose, expected outputs, and ground rules
5. Facilitate: manage energy, time, and equal participation throughout
6. Close with explicitly documented decisions, actions, owners, and due dates

PRACTITIONER TIP

! The facilitator's job is to serve the group's thinking — not to share their own opinions or steer toward a predetermined outcome.

KEY OUTPUTS

- Workshop agenda and facilitation guide
- Session outputs document
- Action log with owners
- Follow-up communication plan

HOW TO MEASURE SUCCESS

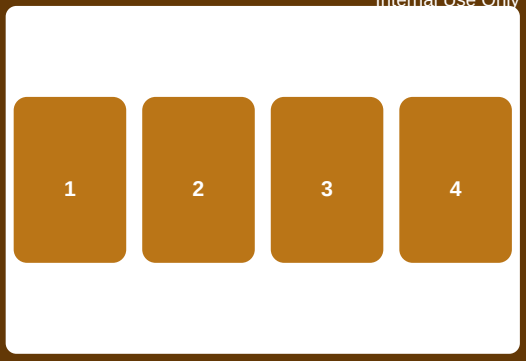
Objective fully achieved; participant engagement and satisfaction score; action completion rate at 30 days

TOOL CATEGORY
People & CI

Internal Use Only

Logic Tree

Break down a complex problem or goal into its component parts using a hierarchical diagram to ensure complete,



WHEN TO USE

For problems where multiple causes or solutions exist and completeness of analysis must be verified before acting.

HOW TO APPLY — STEPS

1. State the problem or objective clearly at the top of the tree
2. Break into 3–5 major branches (MECE: mutually exclusive, collectively exhaustive)
3. Continue branching until actionable root causes or solutions are identified at the bottom
4. Prioritise branches by potential impact and feasibility
5. Action the highest priority items with clear owners and timelines

PRACTITIONER TIP

! MECE is the key discipline: every branch should be distinct, and together they must cover all possibilities — no gaps, no overlaps.

KEY OUTPUTS

- Logic tree diagram
- Prioritised action list
- Tested hypothesis set

HOW TO MEASURE SUCCESS

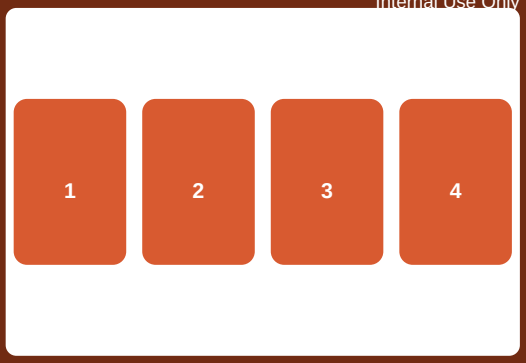
Problem fully resolved with no recurrence; decision confidence rating by stakeholders

TOOL CATEGORY
Operations

Internal Use Only

Multiskilling & Job Rotation

Develop team members' ability to perform multiple roles to reduce single-point dependency, increase operational



WHEN TO USE

When absenteeism creates unacceptable production or service risk; when team engagement is low; when building a high-performing, resilient team.

HOW TO APPLY — STEPS

1. Identify which roles can and should be cross-trained based on operational risk
2. Use the skills matrix to plan logical rotation sequences for each individual
3. Train and rotate systematically on a planned schedule — not only when someone is absent
4. Update the skills matrix after each rotation and sign off competency formally
5. Recognise and reward multi-skill development visibly and consistently

PRACTITIONER TIP

! Rotate regularly, not only in emergencies. Emergency-only rotation means people are never truly competent in their secondary roles.

KEY OUTPUTS

- Job rotation schedule by individual
- Updated and verified skills matrix
- Competency sign-off records for each secondary role

HOW TO MEASURE SUCCESS

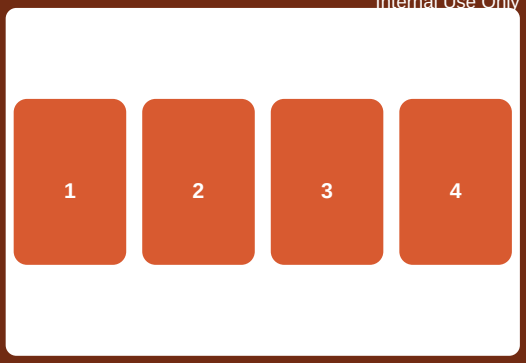
% of team members competent in two or more roles; absence impact on output; employee engagement score

TOOL CATEGORY
People & CI

Internal Use Only

NLC Principles

Apply Natural Leadership Capability principles to develop leaders who enable people to think and perform at their best.



WHEN TO USE

When building leadership development programmes; when team engagement, initiative, or empowerment is consistently low.

HOW TO APPLY — STEPS

1. Assess current leadership behaviours against the NLC capability model
2. Define a target leadership profile relevant to the organisation's context
3. Build an individual development plan for each leader based on the gap
4. Coach leaders using real workplace situations — not classroom exercises alone
5. Measure team outcomes as the primary indicator of leadership effectiveness

PRACTITIONER TIP

! NLC is about what a leader enables in others, not what the leader personally does. Measure the team, not the leader's activity.

KEY OUTPUTS

- Leadership capability assessment
- Individual development plan per leader
- Coaching interaction log
- Team outcome metrics (engagement, initiative, quality)

HOW TO MEASURE SUCCESS

Team engagement score; decisions made successfully below management level; employee-initiated improvement rate

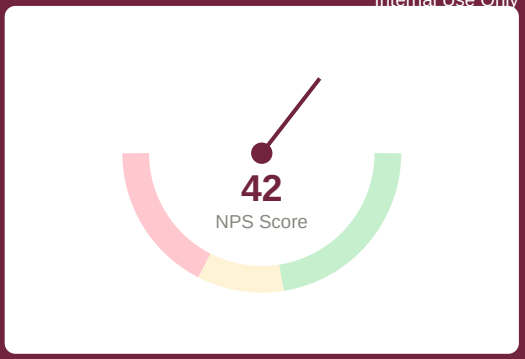
TOOL CATEGORY
People & CI

COMMERCIAL

Internal Use Only

NPS / Survey Tools

Measure customer loyalty and satisfaction using standardised surveys to generate quantitative data for tracking and



WHEN TO USE

Continuously or at defined moments in the customer journey: post-purchase, post-support interaction, or annual relationship review.

HOW TO APPLY — STEPS

1. Select the right survey type for the moment — NPS for loyalty, CSAT for interaction, CES for effort
2. Design short, unambiguous questions — maximum 3 questions per touchpoint
3. Distribute at the right moment in the customer journey — not as an afterthought
4. Analyse overall score AND verbatim comments for themes — the score is a number, the comments contain the insight
5. Act on findings within 30 days and close the loop personally with detractors where possible

PRACTITIONER TIP

! NPS alone tells you almost nothing. The follow-up question 'Why did you give that score?' contains all the actionable information.

KEY OUTPUTS

- NPS, CSAT, or CES score by segment and period
- Verbatim theme analysis report
- Action plan by theme with owners and deadlines

HOW TO MEASURE SUCCESS

NPS trend over rolling 12 months; survey response rate %; score by customer segment; improvement actions closed per quarter

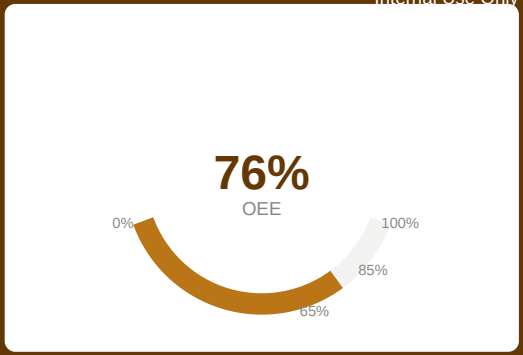
TOOL CATEGORY
Commercial

OPERATIONS

Internal Use Only

OEE (Overall Equipment Effectiveness)

Measure equipment productivity as Availability × Performance × Quality — identifying which of the six big losses to target



WHEN TO USE

When equipment is a constraint on output; to benchmark machine performance and prioritise maintenance investment decisions.

HOW TO APPLY — STEPS

1. Define and measure Availability (planned run time minus unplanned stops)
2. Measure Performance (actual output rate vs theoretical ideal rate)
3. Measure Quality (good parts as a % of total parts produced)
4. Calculate OEE = A × P × Q × 100
5. Identify the largest single loss category and focus improvement there

PRACTITIONER TIP

! World class OEE is 85%+. But first ensure your data definitions are accurate — garbage in equals garbage out on OEE.

KEY OUTPUTS

- OEE dashboard by machine and shift
- Six big losses waterfall analysis
- Improvement action plan with expected OEE impact

HOW TO MEASURE SUCCESS

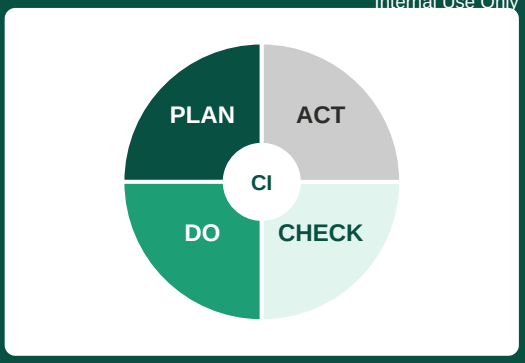
OEE % improvement month on month; reduction in the identified biggest loss category

TOOL CATEGORY
Operations

Internal Use Only

PDCA

A four-step scientific method — Plan, Do, Check, Act — for iterative problem solving and continuous improvement.



WHEN TO USE

For any problem where the root cause or solution is not yet known and needs to be tested empirically.

HOW TO APPLY — STEPS

1. Plan — define problem, root cause hypothesis, and experiment design
2. Do — run a small-scale, controlled test
3. Check — compare results against the hypothesis rigorously
4. Act — standardise if successful; return to Plan with new hypothesis if not

PRACTITIONER TIP

! The Check step is where most teams fail. Be ruthlessly objective — if data doesn't support the hypothesis, the hypothesis was wrong.

KEY OUTPUTS

- PDCA worksheet with hypothesis
- Test results data
- Updated process standard

HOW TO MEASURE SUCCESS

PDCA cycle time in days; % of cycles resulting in sustained, verified improvement

TOOL CATEGORY

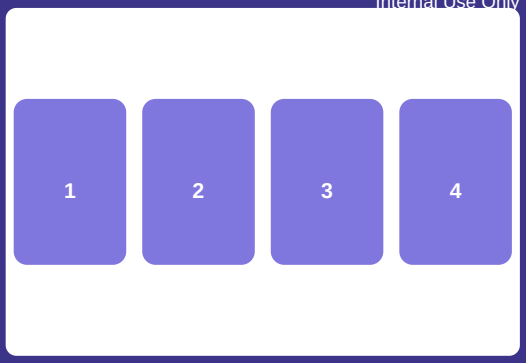
Quality & Process

STRATEGY & PLANNING

Internal Use Only

Project Charter

Formally define a project's scope, objectives, team, timeline, and authority before work begins — creating a shared



WHEN TO USE

Starting any project involving cross-functional resources or lasting more than two weeks.

HOW TO APPLY — STEPS

1. Define problem statement and business case
2. Specify scope — in and out boundaries
3. Set measurable objectives and success criteria
4. Identify team roles, sponsor, and decision rights
5. Get formal sign-off from sponsor before work starts

PRACTITIONER TIP

! A charter that takes longer than two pages usually signals unclear scope. Simplify before starting.

KEY OUTPUTS

- Signed project charter document
- RACI matrix
- High-level milestone timeline

HOW TO MEASURE SUCCESS

On-time delivery; scope change requests count; benefits realised vs projected

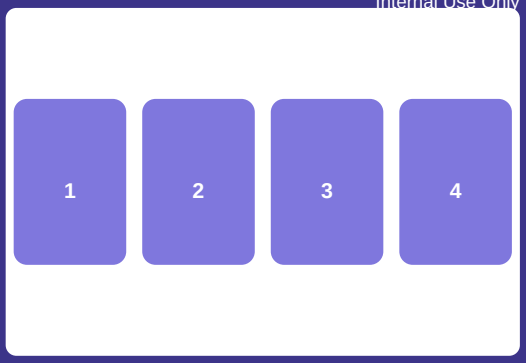
TOOL CATEGORY

Strategy & Planning

STRATEGY & PLANNING

Project Management

Plan, execute, monitor, and close projects on time, within budget, and to the agreed scope and quality standard.



WHEN TO USE

Any initiative with defined deliverables, a timeline, cross-functional dependencies, and resource requirements.

HOW TO APPLY — STEPS

1. Define scope and Work Breakdown Structure
2. Estimate effort and assign resources
3. Build schedule using Gantt chart
4. Manage risks and issues weekly
5. Track progress and communicate status to all stakeholders

PRACTITIONER TIP

Track schedule AND budget weekly. Problems found late are exponentially more expensive to fix.

KEY OUTPUTS

- Project plan and Gantt chart
- Weekly status reports
- Risk and issue register
- Lessons learned log

HOW TO MEASURE SUCCESS

On-time delivery rate; budget variance %; scope change count per project

TOOL CATEGORY

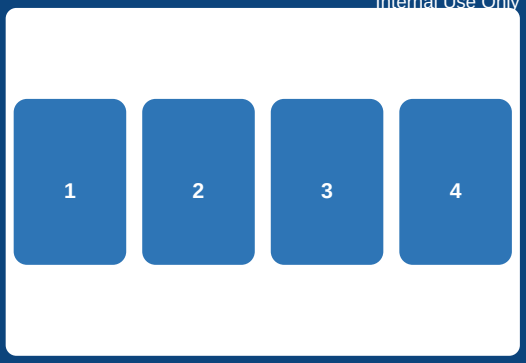
Strategy & Planning

MANAGEMENT SYSTEMS

Internal Use Only

Revision of a Standard

Update existing standards in a controlled, documented way to reflect process improvements, lessons learned, or regulatory



WHEN TO USE

After a process improvement is implemented; when a standard is found to no longer accurately describe what should be done.

HOW TO APPLY — STEPS

1. Identify the specific trigger and reason for the revision
2. Review the change with all people affected by the standard before finalising
3. Update the document with clear version control and date of revision
4. Communicate the change explicitly — do not rely on people noticing a new version number
5. Archive the previous version — never delete previous versions

PRACTITIONER TIP

! Standards that are never revised are either perfect — which is unlikely — or ignored — which is the more likely explanation.

KEY OUTPUTS

- Revised standard with updated version number
- Summary of changes from previous version
- Retraining record for affected personnel

HOW TO MEASURE SUCCESS

Time from improvement to standard update; number of out-of-date standards identified in audit

TOOL CATEGORY

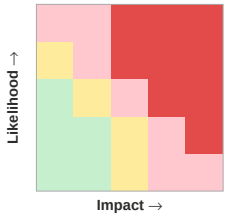
Management Systems

PEOPLE & CI

Internal Use Only

Risk Management

Identify, assess, prioritise, and control risks to prevent losses and protect business continuity — across all departments and



WHEN TO USE

Before major changes, projects, or investments; on a continuous basis for operational, financial, strategic, and safety risks.

HOW TO APPLY — STEPS

1. Identify risks through structured brainstorming, FMEA, historical analysis, and expert input
2. Assess each risk on likelihood and impact using a consistent risk matrix
3. Assign a named risk owner for every risk identified
4. Define control measures and mitigation actions for all high and critical risks
5. Review and update the risk register at least quarterly or after any significant change

PRACTITIONER TIP

! The biggest risks are often the ones everyone assumes 'will never happen here'. Specifically challenge those assumptions first.

KEY OUTPUTS

- Risk register with likelihood/impact scores
- Risk matrix heat map
- Mitigation action plan with owners
- Residual risk summary after controls

HOW TO MEASURE SUCCESS

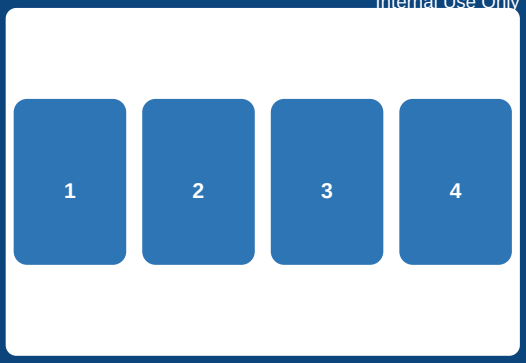
Number of risks identified and effectively controlled; incidents arising from uncontrolled risks; register update frequency

TOOL CATEGORY
People & CI

Internal Use Only

SFM Rules

Define the non-negotiable standards of behaviour and accountability for how a team manages its area — the visible



WHEN TO USE

When establishing a new team management system or when existing standards are inconsistently applied.

HOW TO APPLY — STEPS

1. Identify 5–8 essential observable behaviours for effective team management
2. Document each as a clear, specific, auditable rule
3. Communicate prominently and display at the team board
4. Audit adherence weekly — by the leader and by the team
5. Address non-compliance promptly and consistently without exception

PRACTITIONER TIP

! Rules only stick if the leader models them without exception. One leader ignoring a rule erases months of team compliance.

KEY OUTPUTS

- SFM rules document displayed at board
- Weekly compliance audit checklist
- Compliance trend chart

HOW TO MEASURE SUCCESS

SFM audit compliance %; frequency of rule violations per week; team self-audit accuracy vs management audit

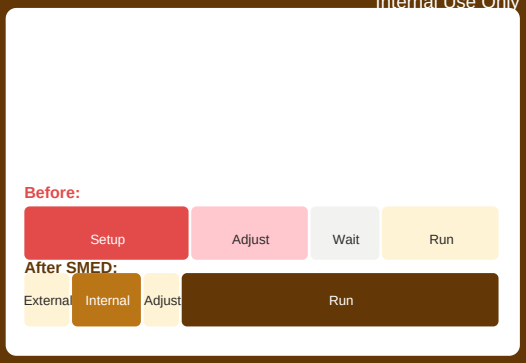
TOOL CATEGORY

Management Systems

OPERATIONS

SMED (Single Minute Exchange of Die)

Reduce equipment changeover time to under 10 minutes by converting internal activities to external and streamlining



WHEN TO USE

When frequent changeovers limit available production time, production flexibility, or batch size reduction.

HOW TO APPLY — STEPS

1. Film the complete current changeover from start to finish on video
2. Separate all tasks into internal (machine stopped) and external (machine running)
3. Convert as many internal steps to external preparation as possible
4. Streamline remaining internal steps using fixtures, quick-release fasteners, and standardisation
5. Train the team and measure every changeover until new time is consistent

PRACTITIONER TIP

! Film the changeover — don't rely on memory or observation. Most waste in changeovers is invisible until you watch the footage repeatedly.

KEY OUTPUTS

- Current vs target changeover time analysis
- Converted external task schedule
- Standard changeover procedure with photos

HOW TO MEASURE SUCCESS

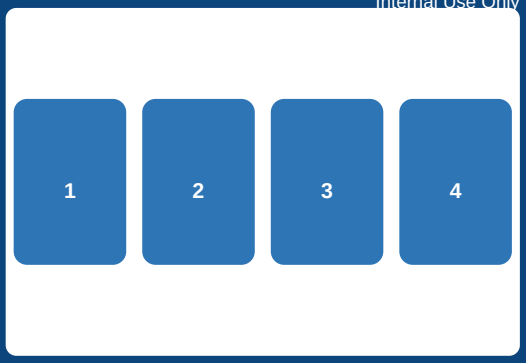
Changeover time reduction % vs baseline; additional production capacity released; changeover frequency enabled

TOOL CATEGORY
Operations

Internal Use Only

SMF Management

Implement a structured daily, weekly, and monthly management framework using standard meetings and visual



WHEN TO USE

When teams operate inconsistently; when management time is dominated by fire-fighting instead of prevention.

HOW TO APPLY — STEPS

1. Design the meeting and reporting cadence across all management levels
2. Define the KPIs, visual boards, and escalation triggers for each level
3. Train team leaders in facilitating structured daily meetings
4. Audit compliance and quality of the SMF in the first 90 days
5. Refine based on what adds value and remove what does not

PRACTITIONER TIP

! The SMF only works if leaders attend consistently and act visibly on what they see. Passive attendance destroys credibility.

KEY OUTPUTS

- SMF blueprint with meeting standards
- Visual board design for each level
- Cadence calendar
- 90-day audit results

HOW TO MEASURE SUCCESS

Meeting compliance rate %; time from problem identified to escalation; issue resolution cycle time

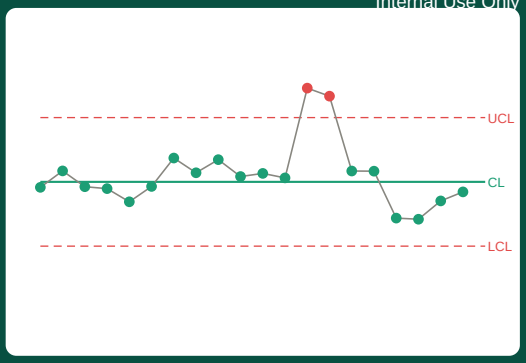
TOOL CATEGORY

Management Systems

Internal Use Only

SPC

Apply Statistical Process Control charts to distinguish common cause variation (normal) from special causes requiring



WHEN TO USE

In any repetitive process where consistency matters — production, service delivery, data processing, clinical care.

HOW TO APPLY — STEPS

1. Select the quality characteristic to monitor
2. Choose the appropriate chart type (X-bar R, p-chart, u-chart, etc.)
3. Calculate control limits from baseline stable-process data
4. Plot new data points in real time as the process runs
5. React to special cause signals only — investigate and eliminate them

PRACTITIONER TIP

! Never tamper with a process that is in statistical control. Tampering only adds variation and makes things worse.

KEY OUTPUTS

- Control charts with calculated limits
- Process capability indices (Cp, Cpk)
- Special cause event log and investigation log

HOW TO MEASURE SUCCESS

Cpk trend over time; number of special cause events per month; false alarm rate

TOOL CATEGORY

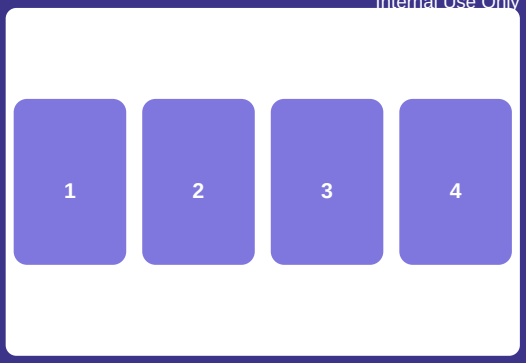
Quality & Process

STRATEGY & PLANNING

Internal Use Only

SWOT Analysis

Identify internal Strengths and Weaknesses alongside external Opportunities and Threats to inform strategic decisions.



WHEN TO USE

Before launching a new product, entering a new market, or facing a significant competitive challenge.

HOW TO APPLY — STEPS

1. Gather cross-functional team (max 8 people)
2. Brainstorm each quadrant in silence first
3. Cluster and prioritise items
4. Identify strategic options from SO/ST/WO/WT intersections
5. Document and share findings with leadership

PRACTITIONER TIP

! Be brutally honest about weaknesses. A SWOT with no weaknesses is a SWOT that won't help you.

KEY OUTPUTS

- SWOT matrix
- Strategic options list
- Action priorities

HOW TO MEASURE SUCCESS

Number of strategic options generated; % actioned within 90 days

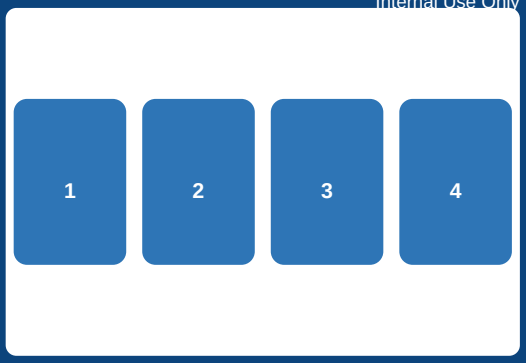
TOOL CATEGORY

Strategy & Planning

Internal Use Only

Shift Handover

Ensure complete, structured transfer of critical information between shifts so incoming teams start without delay, rework,



WHEN TO USE

Wherever shift work exists — and in any role with formal handovers: clinical, IT on-call, project management.

HOW TO APPLY — STEPS

1. Define a standard handover template covering KPIs, open issues, safety, priorities, and equipment status
2. Allocate a fixed 10–15 minutes for every handover — never skip it
3. Outgoing team owns completeness and clarity of the handover
4. Incoming team confirms understanding and asks questions before the handover ends
5. Log all open items and track them to closure

PRACTITIONER TIP

! A handover where the outgoing team says 'it's all fine' and leaves in two minutes is not a handover — it's a gap in accountability.

KEY OUTPUTS

- Completed handover log
- Open issues tracker with owners
- Handover compliance checklist

HOW TO MEASURE SUCCESS

Issues arising from poor or missed handovers per month; handover duration compliance; start-up delay reduction

TOOL CATEGORY

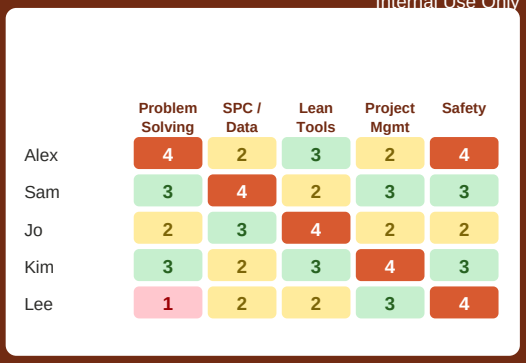
Management Systems

PEOPLE & CI

Internal Use Only

Skills Matrix

Visualise the skill levels of each team member across all required competencies to identify gaps, single-point



	Problem Solving	SPC / Data	Lean Tools	Project Mgmt	Safety
Alex	4	2	3	2	4
Sam	3	4	2	3	3
Jo	2	3	4	2	2
Kim	3	2	3	4	3
Lee	1	2	2	3	4

WHEN TO USE

During team formation; annual performance and development cycles; before cross-training; when planning cover for absences or growth.

HOW TO APPLY — STEPS

1. Define all required competencies for the team with observable, assessable descriptions
2. Assess each team member using a consistent four-level scale (0–4)
3. Display the matrix visibly in the team area — not hidden in a file
4. Identify critical gaps: competencies with only one person capable, or zero coverage
5. Build individual development plans and a team training calendar to close the priority gaps

PRACTITIONER TIP

! Only include competencies you can actually verify through observation or testing. A matrix of unverifiable claims is not a skills matrix.

KEY OUTPUTS

- Skills matrix chart displayed in the team area
- Critical gap analysis
- Individual development plan for each team member
- Training calendar and competency sign-off records

HOW TO MEASURE SUCCESS

% of critical competencies with two or more people fully competent; average team skill level growth per quarter

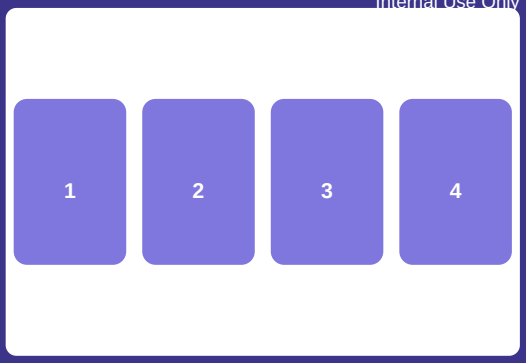
TOOL CATEGORY

People & CI

STRATEGY & PLANNING

Stakeholder Identification

Map all people and groups affected by or able to influence a project, then plan engagement to maximise support and



WHEN TO USE

At the start of any project or change initiative — before decisions are made that affect people.

HOW TO APPLY — STEPS

1. List all potential stakeholders
2. Rate each on Power vs Interest grid
3. Define engagement approach per quadrant
4. Assign relationship owner for each key stakeholder
5. Review and update monthly

PRACTITIONER TIP

Include resistors — ignoring them doesn't make them go away. The hardest stakeholders need the most attention.

KEY OUTPUTS

- Stakeholder map
- Engagement plan
- Communication calendar

HOW TO MEASURE SUCCESS

Stakeholder satisfaction score; % of key stakeholders engaged on schedule

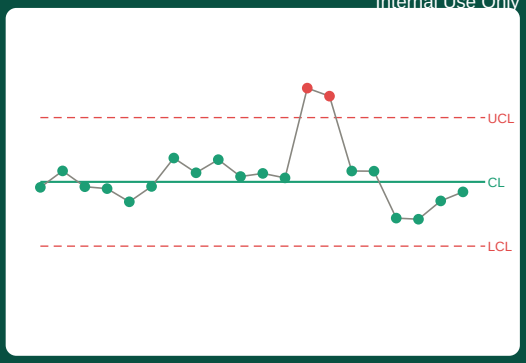
TOOL CATEGORY

Strategy & Planning

QUALITY & PROCESS

Statistical Process Control

Monitor process outputs using control charts to detect changes in process behaviour early — before they produce



WHEN TO USE

In any repetitive process where early detection of change enables prevention rather than detection and rework.

HOW TO APPLY — STEPS

1. Identify the critical-to-quality characteristic
2. Select the appropriate chart type for the data
3. Establish control limits from a period of stable process operation
4. Plot results in real time as the process runs
5. React to signals (points outside limits, runs, trends) with structured investigation

PRACTITIONER TIP

Control limits come from process data. Specification limits come from customer requirements. They are NEVER the same thing.

KEY OUTPUTS

- Real-time control charts
- Process capability summary
- Reaction plan for each signal type

HOW TO MEASURE SUCCESS

False alarm rate; time to detect a genuine process shift; Cpk improvement trend

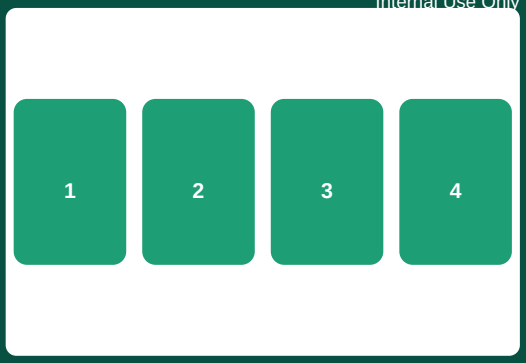
TOOL CATEGORY

Quality & Process

Internal Use Only

Stratification Diagram

Separate aggregate data into meaningful subgroups — by shift, machine, product, or operator — to reveal hidden



WHEN TO USE

When aggregate data masks the true source of a problem — 'the average looks okay but something is clearly wrong'.

HOW TO APPLY — STEPS

1. Define the problem metric to investigate
2. Identify potential stratification factors (shift, machine, operator, etc.)
3. Collect data tagged by each factor
4. Plot each stratum separately on a bar or line chart
5. Focus on the stratum with the worst performance and investigate

PRACTITIONER TIP

! The best stratification factor is the one that creates the biggest difference between groups — that's where to look.

KEY OUTPUTS

- Stratified comparison charts
- Performance table by stratum
- Root cause hypothesis for worst stratum

HOW TO MEASURE SUCCESS

Variance explained by the stratification factor; defect rate in worst-performing stratum

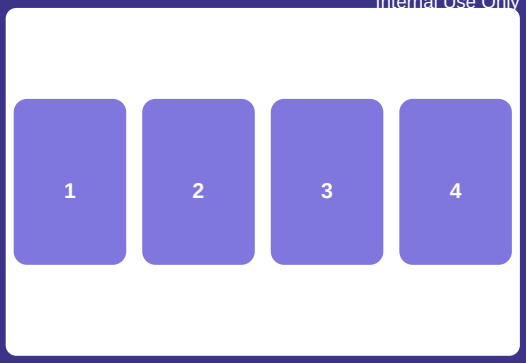
TOOL CATEGORY

Quality & Process

STRATEGY & PLANNING

Tactical Implementation Plan

Break a strategic objective into 90-day action plans with clear owners, milestones, and resource allocations.



WHEN TO USE

After setting annual targets — to translate strategy into specific, executable actions at team level.

HOW TO APPLY — STEPS

1. Define the objective and success metric
2. Identify all actions required to achieve it
3. Assign owner, start date, end date, and resources to each
4. Review progress weekly in team meetings
5. Update plan at each monthly review cycle

PRACTITIONER TIP

Build the plan WITH the people who will execute it — not FOR them. Ownership starts at creation.

KEY OUTPUTS

- 90-day action plan
- Milestone tracker
- Resource plan

HOW TO MEASURE SUCCESS

Milestone on-time completion rate; strategic objective achievement rate

TOOL CATEGORY

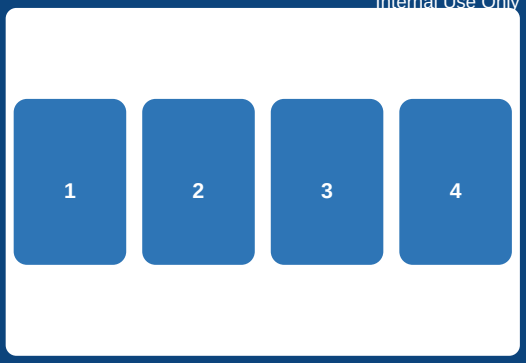
Strategy & Planning

MANAGEMENT SYSTEMS

Internal Use Only

Training of a Standard

Deliver structured, verified training on a work standard to ensure consistent application by all relevant team members.



WHEN TO USE

When onboarding new team members; when a standard has been revised; when deviations are found in practice.

HOW TO APPLY — STEPS

1. Confirm the standard is current, correct, and approved before training begins
2. Prepare training materials including live demonstration — not document only
3. Deliver training using the sequence: tell, show, observe, confirm understanding
4. Require the trainee to demonstrate the standard independently before sign-off
5. Record training completion in the team skills matrix and training register

PRACTITIONER TIP

! Training is only complete when the trainee CAN demonstrate the standard alone — not when they have sat through a presentation about it.

KEY OUTPUTS

- Training completion record signed by trainer and trainee
- Updated skills matrix
- Post-training competency assessment result

HOW TO MEASURE SUCCESS

Training completion rate %; time to demonstrated competency; post-training deviation rate at 30 days

TOOL CATEGORY

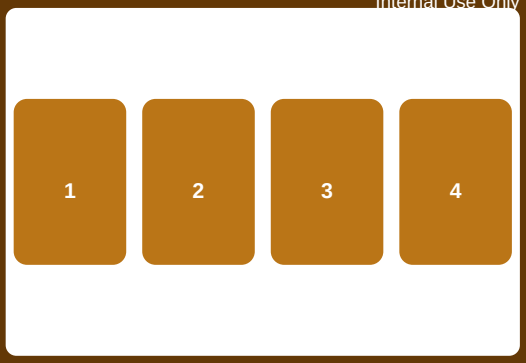
Management Systems

OPERATIONS

Internal Use Only

VSM Data Capture

Capture accurate process data during a Value Stream Mapping gemba walk — cycle times, changeover times,



WHEN TO USE

During the current state VSM gemba walk. Walk the process with this sheet, measure actual times (minimum 5 cycles per step), and record everything you observe.

HOW TO APPLY — STEPS

1. Define the value stream boundaries: customer demand, start point, and end point
2. Calculate takt time: available seconds per day divided by customer demand per day
3. Walk the process from customer back to raw material — observe, do not manage
4. At each step: measure actual cycle time (5 cycles minimum), C/O time, and uptime %
5. Count physical inventory between each step and estimate queue time in hours
6. Note value-added (Y) vs non-value-added (N) classification for each step
7. Record number of operators and any waste or problem observations
8. Calculate process efficiency: total VA time divided by total lead time

PRACTITIONER TIP

! NEVER estimate — measure. Walk upstream from customer to supplier. Use a stopwatch. The data you collect here becomes the foundation for every improvement decision on the VSM.

KEY OUTPUTS

- Current state process data table (CT, C/O, uptime, inventory, queue time)
- Process efficiency % and lead time calculation
- Value-added vs non-value-added step classification
- Waste and problem observation notes for each step
- Input data for drawing the current state VSM

HOW TO MEASURE SUCCESS

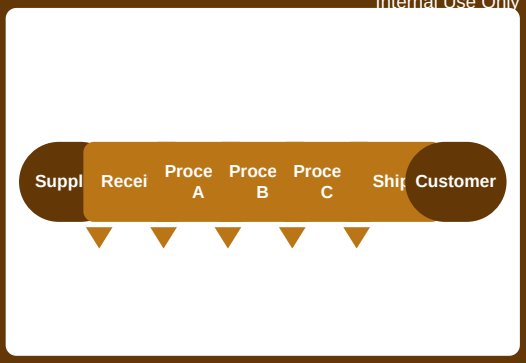
Total lead time (days); Value-added time (seconds); Process efficiency %; Total WIP inventory (units); % steps operating below takt time

TOOL CATEGORY
Operations

Internal Use Only

Value Stream Mapping

Visualise the entire flow of material and information from raw material to customer — revealing waste, delays, and



WHEN TO USE

Before any major process improvement initiative; whenever lead time is the primary constraint or complaint.

HOW TO APPLY — STEPS

1. Select a specific product family or value stream to map
2. Walk the process end-to-end with a pencil and stopwatch — never map from memory
3. Record process metrics at each step: cycle time, lead time, inventory levels, uptime
4. Identify the largest sources of waste and delay
5. Design the future state map with specific improvements
6. Build a phased implementation roadmap to close the gap

PRACTITIONER TIP

! A VSM done at a desk using estimates is fiction. Walk the actual process with real data every single time.

KEY OUTPUTS

- Current state value stream map
- Future state value stream map
- Implementation roadmap
- Lead time and value-added ratio analysis

HOW TO MEASURE SUCCESS

Lead time reduction %; value-added ratio improvement; total WIP inventory reduction

TOOL CATEGORY

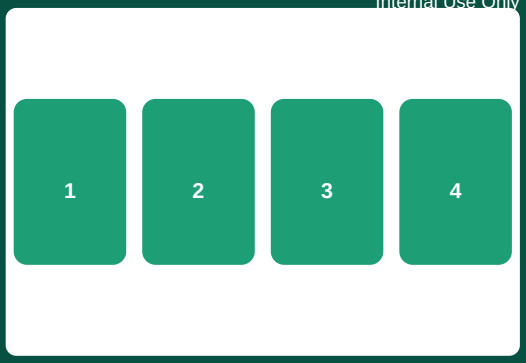
Operations

QUALITY & PROCESS

Internal Use Only

Waste Identification

Identify and quantify the 8 wastes (TIMWOODS) — Transport, Inventory, Motion, Waiting, Overproduction, Over-processing,



WHEN TO USE

During gemba walks, process reviews, or when looking for cost reduction and lead time improvement opportunities.

HOW TO APPLY — STEPS

1. Train the team on all 8 waste types with real examples
2. Walk the process physically — never observe from a desk
3. Record each waste observed with frequency and estimated cost impact
4. Prioritise the top three wastes by total impact
5. Launch targeted improvement actions for each priority waste

PRACTITIONER TIP

! The 8th waste — unused human talent and creativity — is the most expensive and most overlooked waste in organisations.

KEY OUTPUTS

- Waste observation sheet
- Estimated waste cost by type
- Improvement priority list

HOW TO MEASURE SUCCESS

Estimated waste cost eliminated; process lead time reduction %; value-added ratio before and after

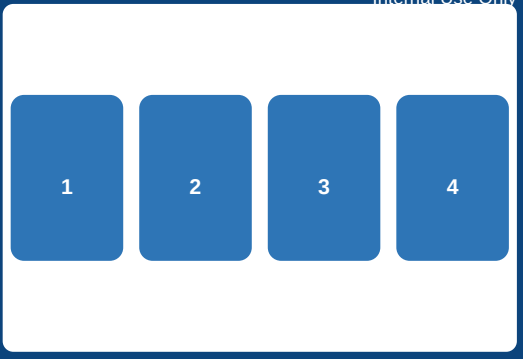
TOOL CATEGORY

Quality & Process

Internal Use Only

Work Instructions Guidelines

Provide a framework for writing work instructions that are clear, visual, concise, and actually used by operators at the



WHEN TO USE

When creating or revising any written procedure, work instruction, or operating standard.

HOW TO APPLY — STEPS

1. Write the instruction WITH the people who perform the task — not about them
2. Use photographs, diagrams, and numbered steps — minimise text where possible
3. Limit to one page where possible; two pages maximum for complex tasks
4. Test with an inexperienced operator: if they can follow it alone, it is good enough
5. Post at the point of use where it is visible during work — not in a folder

PRACTITIONER TIP

! If explaining a single step requires a paragraph, that step is too complex and needs to be broken down further.

KEY OUTPUTS

- One or two page visual work instruction
- Photo-based standard at point of use
- Validation result from test with new operator

HOW TO MEASURE SUCCESS

Work instruction adherence rate %; errors per instruction cycle; proportion of instructions that are one page

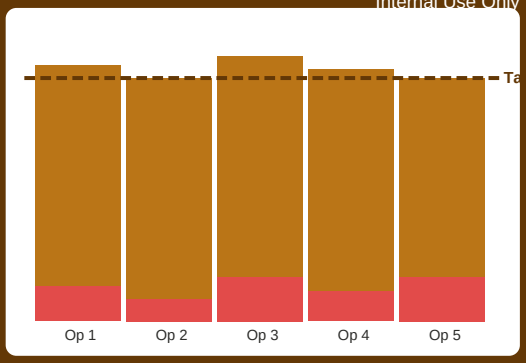
TOOL CATEGORY

Management Systems

OPERATIONS

Yamazumi

A stacked bar chart visualising individual operator cycle times vs takt time — enabling balanced work redistribution to



WHEN TO USE

When designing or rebalancing a production line, service cell, or office workflow with multiple operators.

HOW TO APPLY — STEPS

1. Measure time required for every individual task each operator performs
2. Build stacked bars — one per operator — total bar height equals their cycle time
3. Draw the takt time as a horizontal reference line across all operators
4. Identify operators over takt (overloaded) and under takt (waiting)
5. Redistribute tasks collaboratively with operators to balance the line

PRACTITIONER TIP

! Rebalance WITH the operators — they know which tasks can be combined, split, or relocated in the flow.

KEY OUTPUTS

- Yamazumi stacked bar chart
- Rebalanced work assignment table
- Updated operator standard work sheets

HOW TO MEASURE SUCCESS

Line balance rate %; average operator utilisation vs takt; headcount efficiency improvement

TOOL CATEGORY
Operations